

20EC503 – MI

Max. Marks: 100

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Marks	BT*	CO*	PO*
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1.
 - a) Mention any four addressing modes of 8051 microcontroller. Using two examples for each explain addressing mode.
 - b) Write an Assembly Language Program to convert six 2-digit BCD numbers stored in memory location 20H onwards to Hexadecimal. Store the result of conversion in memory location 50H onwards. Perform addition of converted numbers and store the LSB of result in Register A and MSB in R3. Use Subroutine function for number conversion. Take an example to illustrate the output of the program. Also include comments in the program.
2.
 - a) With the neat diagram, explain 128 bytes RAM memory organization of 8051 microcontroller with address in detail. Give any two instructions as an example of bit addressable programming.
 - b) Draw a neat architecture diagram of 8051 microcontroller. With an example for each, explain PUSH and POP instruction of 8051 microcontroller. Analyze the Program given below. What is the data stored in SP, R0, R1, 30H, 31H and R5 at the end of execution?

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ORG 00H
MOV A, #120
PUSH 0E0H
MOV B, #53H
MOV 12H, #25H
MUL AB
PUSH 0E0H
PUSH 0F0H
MOV A, #32H
ADD A, #14H
DA A
PUSH 0E0H
MOV R0, #12H
MOV A, @R0
RR A
MOV R1,A
POP 02H
MOV 30H,R2
POP 03H
MOV 31H,R3
POP 05H
END

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12 L3 1 1

3. a) With an interfacing diagram, write an assembly language program to display number '1234' on a multiplexed seven segment display using only 12 I/O port pins.

8 L3 2 1

- b) Draw a neat block diagram of Timer 1 in Mode1 of 8051 microcontroller. Write the steps to program timers in Mode 1. Write an assembly language program to generate a square wave of 2kHz frequency on port pin P1.5 of 8051 microcontroller. Assume crystal frequency as 11.0592kHz. Also include comments in the program. 12 L3 2 1
4. a) Draw the internal port structure of 8051 microcontroller. Assume that two switches S1 and S2 are connected to P3.2 and P3.3 of 8051 microcontroller. When S1 is closed microcontroller must perform up count once and when S2 is closed microcontroller must perform down count once. Eight LEDs are connected to port 0 to display the count. Draw the interfacing diagram. Write an 8051 assembly code to illustrate the counter. 8 L3 2 1
- b) Give the bit description for TMOD and SCON register of 8051 microcontroller. Assume that 8051 port is connected to the COM port of the PC to send and receive data serially. P1 and P2 of the 8051 microcontroller are connected to LEDs and switches, respectively. Write an 8051 program which must include 3 parts as follows
 (i) Send the message "We Are Ready"
 (ii) receive any data sent by the PC and put it on LEDs connected to P1
 (iii) get data on switches connected to P2 and send it to the PC serially
 The program should perform part (i) once, but parts (ii) and (iii) continuously. Use 4800 baud rate. Also include comments in the program. 12 L3 2 1
- Unit – III**
5. a) Describe the memory map of MSP430 microcontroller. With the help of instructions, explain how the data is copied to the stack and copied from the stack in MSP430. 10 L2 3 1
- b) Draw the neat functional block diagram of MSP430. Explain the CPU and Registers of MSP430 in detail. 10 L2 3 1
6. a) Write a short note on Interrupts in MSP430. Also explain what happens when an interrupt is requested? 10 L2 3 1
- b) Explain sending data to a Slave device and reading data from a Slave device using I2C bus in MSP430 with the help of timing diagram. 10 L2 3 1
- Unit – IV**
7. a) Explain the structure of 5 stage pipeline organization of ARM processor in detail. 10 L2 4 1
- b) What are exceptions in ARM and how these exceptions are handled in ARM? With an example for each, illustrate how Branch and Branch with link instruction works in ARM. 10 L2 4 1
8. a) Explain the basic operating modes of ARM processor. Also list and explain features of ARM processor. 10 L2 4 1
- b) Mention the ARM instruction format. Explain the role of condition field in detail. With an example explain Load and store instruction of ARM processor. 10 L2 4 1
- Unit – V**
9. a) Draw the Multiplier architecture of ARM and explain multiplier design with an example. 10 L2 5 1
- b) Explain the principle of cross bar switch Barrel shifter. 10 L2 5 1

10. a) Draw and briefly explain the ALU structure of ARM6. Also draw the ARM2 ALU logic and write any four ARM2 ALU function codes for one result bit.
- b) With the help of logic diagram, explain the concept of Carry Look ahead scheme used in ARM2 and Carry select adder used in ARM6.

10	L2	5	1
10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
